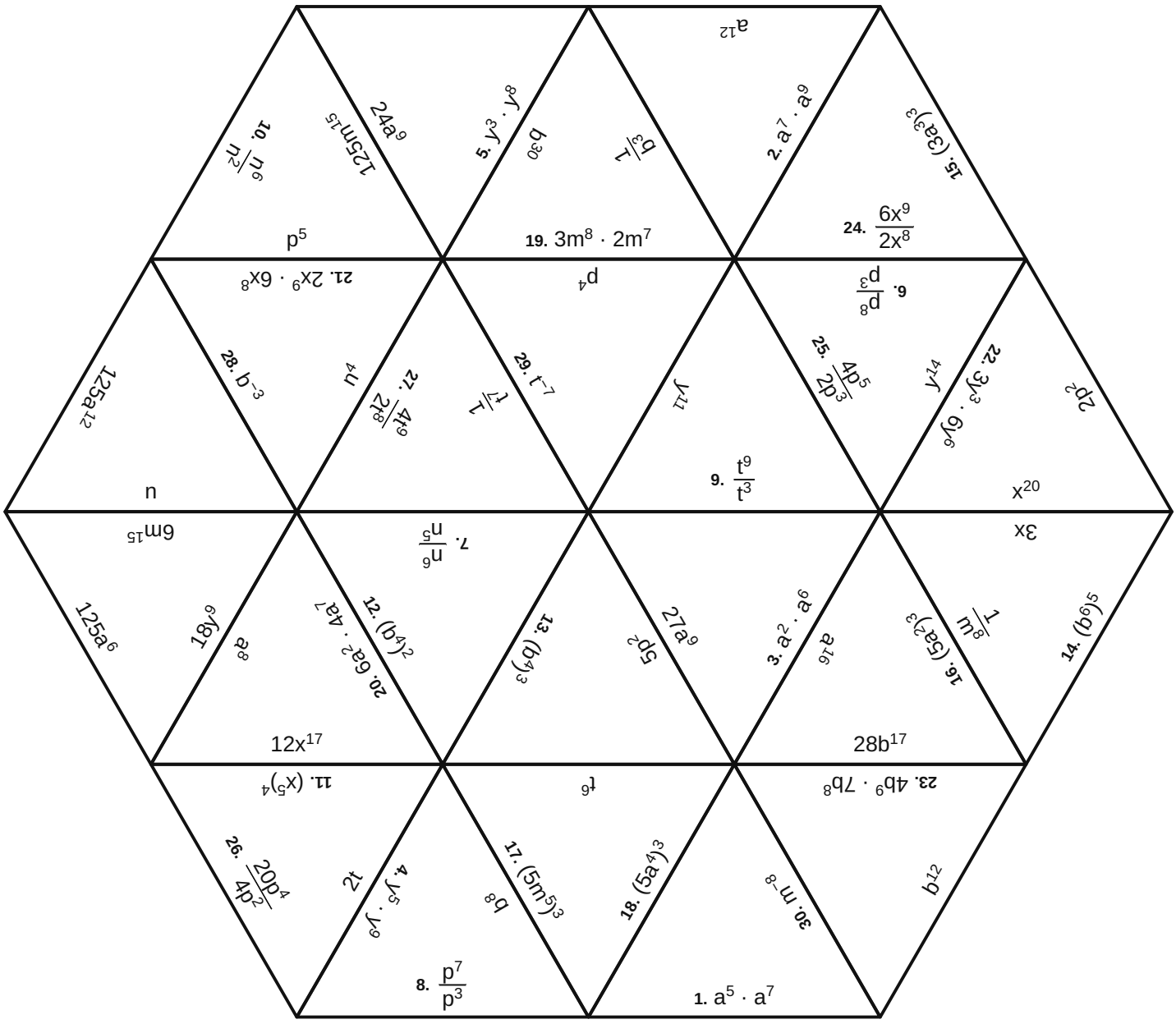
Exponent Rules Tarsia

Mr. K's Classroom · Hexagon of 24 triangles, 30 matches Name: _____

Directions: Simplify each expression. Write answers with positive exponents. Cut out the 24 triangles. Match each problem to its answer so that touching edges fit, and build a **hexagon**. Glue it on a sheet of paper. Show your work for every problem on the work pages (the number before each problem tells you where).



Show Your Work

Exponent Rules Tarsia · Problems 1–15 Name: _____

1. $a^5 \cdot a^7$	**2.** $a^7 \cdot a^9$	**3.** $a^2 \cdot a^6$
4. $y^5 \cdot y^9$	**5.** $y^3 \cdot y^8$	**6.** $\frac{p^8}{p^3}$
7. $\frac{n^6}{n^5}$	**8.** $\frac{p^7}{p^3}$	**9.** $\frac{t^9}{t^3}$
10. $\frac{n^6}{n^2}$	**11.** $(x^5)^4$	**12.** $(b^4)^2$
13. $(b^4)^3$	**14.** $(b^6)^5$	**15.** $(3a^3)^3$

Show Your Work

Exponent Rules Tarsia · Problems 16–30 Name: _____

16. $(5a^2)^3$	**17.** $(5m^5)^3$	**18.** $(5a^4)^3$
19. $3m^8 \cdot 2m^7$	**20.** $6a^2 \cdot 4a^7$	**21.** $2x^9 \cdot 6x^8$
22. $3y^3 \cdot 6y^6$	**23.** $4b^9 \cdot 7b^8$	**24.** $\frac{6x^9}{2x^8}$
25. $\frac{4p^5}{2p^3}$	**26.** $\frac{20p^4}{4p^2}$	**27.** $\frac{4t^9}{2t^8}$
28. b^{-3}	**29.** t^{-7}	**30.** m^{-8}